

SQL Aggregate Functions

Easy Notes & Quick Reference Guide — MAX, MIN, COUNT, SUM, AVG

Aggregate functions operate on a set of rows and return a single calculated summary result.

SAMPLE DATASET: `EMP` TABLE

Eid	Ename	Dept	Salary
1	Ravi	HR	10000
2	Varun	IT	20000
3	Nitin	HR	30000
4	Robin	IT	30000
5	Amey	Sales	50000
6	Amit	Sales	NULL

1. MAX() & 2. MIN()

MAX(column)

Output: 50000

Finds the maximum (highest) value in the set. Ignores `NULL` values (does not treat them as 0).

```
SELECT MAX(salary) FROM Emp;
```

MIN(column)

Output: 10000

Finds the minimum (lowest) value in the set.

```
SELECT MIN(salary) FROM Emp;
```

3. COUNT() VARIATIONS

★ Key Exam Distinction for COUNT:

- `COUNT(*)` → Counts **total rows** (including `NULL`s).
- `COUNT(column)` → Counts **non-NULL values** only.
- `COUNT(DISTINCT column)` → Counts **unique non-NULL values**.

COUNT(*)

Output: 6

Counts all 6 rows in the `Emp` table.

```
SELECT COUNT(*) FROM Emp;
```

COUNT(salary)

Output: 5

Counts non-NULL salary entries (excludes Amit's `NULL` salary).

3. COUNT() (CONTINUED)

COUNT(DISTINCT salary)

Output: 4

Distinct non-NULL values are 10000, 20000, 30000, 50000 (30000 appears once, NULL is ignored).

```
SELECT COUNT(DISTINCT salary) FROM Emp;
```

4. SUM()

SUM(salary)

Output: 140000

Calculation: $10000 + 20000 + 30000 + 30000 + 50000 = 140000$

```
SELECT SUM(salary) FROM Emp;
```

SUM(DISTINCT salary)

Output: 110000

Calculation: $10000 + 20000 + 30000 + 50000 = 110000$

```
SELECT SUM(DISTINCT salary) FROM Emp;
```

5. AVG()

AVG(salary)

Output: 28000

Formula: $\text{SUM} / \text{COUNT}(\text{non-NULL values}) \rightarrow 140000 / 5 = 28000$

```
SELECT AVG(salary) FROM Emp;
```

AVG(DISTINCT salary)

Output: 27500

Formula: $\text{SUM}(\text{DISTINCT}) / \text{COUNT}(\text{DISTINCT}) \rightarrow 110000 / 4 = 27500$

```
SELECT AVG(DISTINCT salary) FROM Emp;
```

QUICK REVISION TABLE

Function	Purpose & Description
MAX()	Returns highest value
MIN()	Returns lowest value
COUNT(*)	Returns total row count
COUNT(column)	Returns non-NULL value count
COUNT(DISTINCT column)	Returns unique non-NULL value count
SUM()	Returns total addition of values